

KOMAL SAI JAKKA

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Education

Sardar Vallabhbhai National Institute of Technology, Surat

Aug 2024– May 2028(Expected)

Bachelor of Technology in Artificial Intelligence

CGPA: 8.12(Till Sem 4)

Projects

Topology-Aware Sepsis Prediction (CAT³P) | *Python, PyTorch, DuckDB, MIMIC-IV*

Jan 2026

- Developed a deep learning framework for four-stage sepsis severity prediction (No Sepsis, Sepsis, Severe Sepsis, Septic Shock) using temporal electronic health records from the MIMIC-IV dataset.
- Designed a topology-aware Graph Neural Network (CAT³P) to model interactions among organ systems while capturing temporal disease progression.
- Built an end-to-end preprocessing pipeline including Sepsis-3 label generation, hourly time-series alignment, missing-value imputation, feature normalization, and graph tensor construction.
- Improved minority-class prediction using class-weighted cross-entropy loss and weighted random sampling, achieving **57.8% Macro F1-score** and **0.8869 Macro AUROC**.

MindSettler - Mental Wellness Session Booking Platform | *HTML, CSS, JavaScript, Django*

DEC 2025

- Developed a calm and user-friendly UI/UX design to enhance user experience and accessibility.
- Built a responsive web platform that enables users to book online and offline therapy sessions with real-time scheduling functionality.
- Ensured smooth frontend-backend integration for efficient data flow and booking management.
- Implemented Django backend logic to handle session bookings, user data management, and form submissions.

MediCare - Hospital Management System | *HTML, CSS, JavaScript, Django*

APRIL 2024

- Implemented secure authentication and role-based access control to ensure data privacy and controlled system access.
- Designed responsive user interfaces to improve usability for patients, doctors, and administrators.
- Integrated database operations for efficient storage and retrieval of medical and appointment data.
- Developed a full-stack hospital management web application using Django for backend and HTML, CSS, JavaScript for frontend to manage patient records, doctor information, and appointment scheduling.

Technical Skills

Programming Languages: Python, C++, HTML, CSS, JavaScript, SQL, C, Pandas, Numpy

Technologies/Frameworks: Django*, MySQL

Skills: Data Structures & Algorithms, Database Management System (DBMS), Object-Oriented

Programming (OOPs), Design & Analysis of Algorithms(DAA), Artificial Intelligence, Machine Learning

Extracurricular

- Codeforces Rating: **878** with participation in competitive programming contests.
- Practiced and solved **175+ algorithmic problems** on LeetCode, enhancing logical thinking, coding efficiency, and time complexity optimization skills.
- Won **1st Prize** in Squidathon, a high-intensity team competition involving creative meme challenges, game-based problem-solving rounds, and real-world task execution under time constraints.